P.L.OAVIKSHA(192572001)

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1715**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

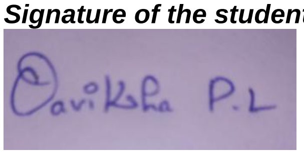
Annexure A

ARTICLE REVIEW

Code of Conduct:

I P.L.OAVIKSHA(192572001) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

TASK 1: ARTICLE SUMMARY

(a) Full Citation, Domain/Application Area and ML Problem

Authors: Elena Ballante, Marta Galvani, Pierpaolo Uberti, and Silvia Figini

Title: *Polarized Classification Tree Models: Theory and Computational Aspects*

Journal: *Journal of Classification*

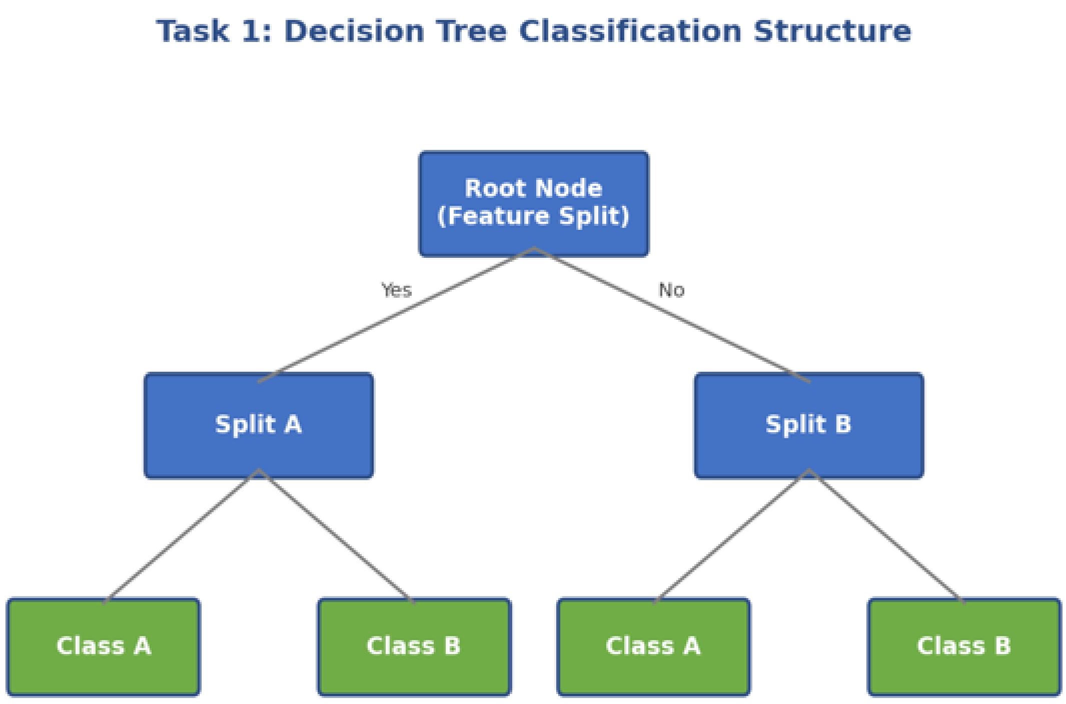
Year: 2021

Volume: 38, Issue 3, Pages 481–499

DOI: 10.1007/s00357-021-09383-8

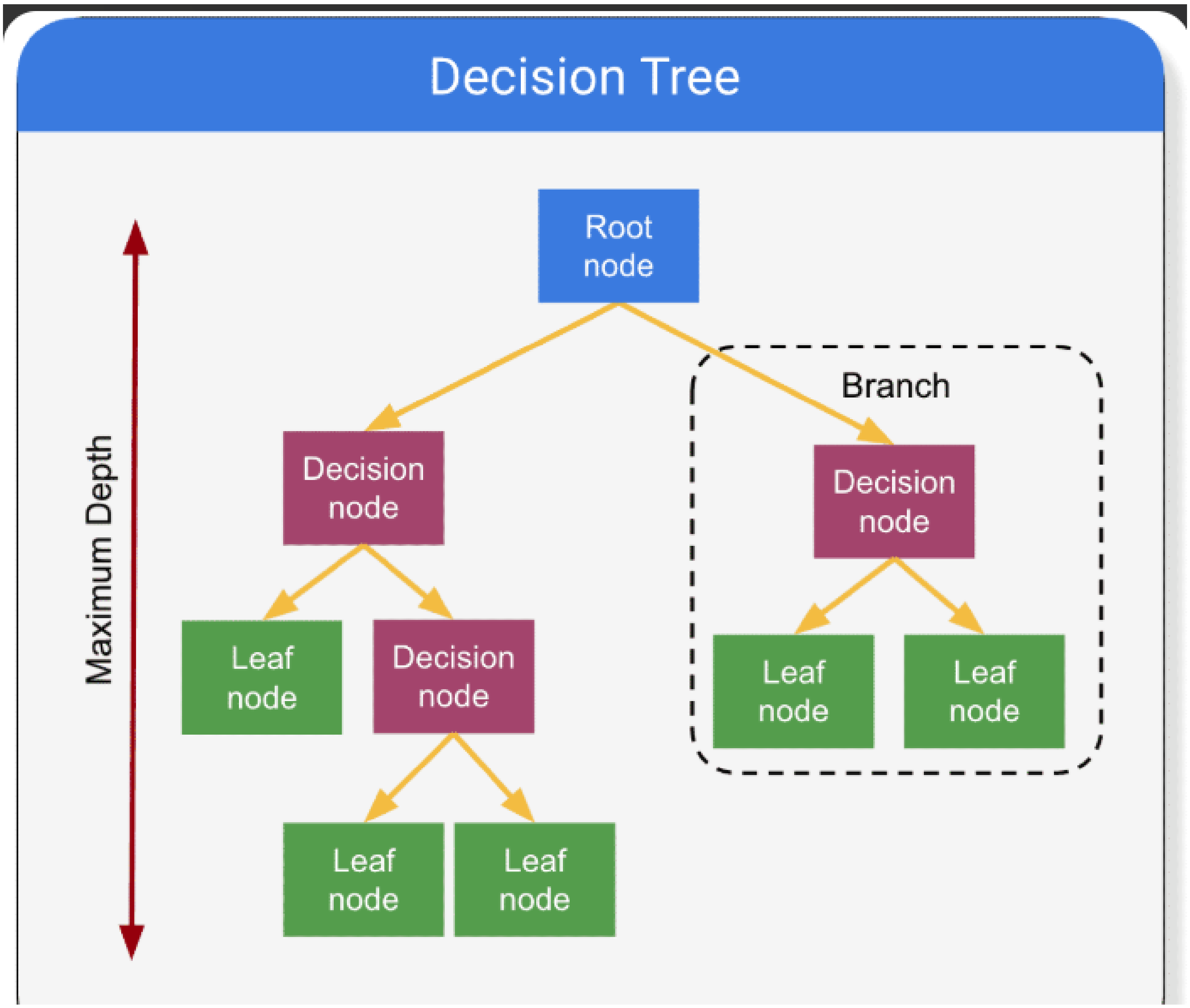
Domain/Application Area:
Machine Learning, Classification, and Decision Tree Modelling.

Specific ML Problem Addressed:
The article addresses the problem of selecting effective splitting criteria in classification trees. Traditional methods such as the **Gini Index** and **Information Gain** mainly focus on node impurity and do not sufficiently consider the distribution of the input variables within the nodes.



(b) Article Summary

The article proposes a **Polarized Classification Tree (PCT)** model to improve the splitting process in classification trees. Traditional Decision Tree methods commonly use the Gini Index and Information Gain to select the best split. The authors identify a research gap in these methods because they mainly measure the impurity of the resulting nodes and do not sufficiently consider the distribution of the input variables. To address this limitation, the authors introduce a new **polarization index** for measuring the quality of a split. The proposed measure considers the variability between and within groups as well as the distribution of observations. The authors also develop an algorithm that incorporates this measure into the growth of a classification tree. The proposed approach is evaluated using simulation experiments and real datasets from the UCI repository. The simulation results indicate that the proposed method can outperform commonly used impurity measures. Experiments on real datasets show that Polarized Classification Trees are competitive with, and sometimes better than, traditional Decision Trees based on Gini Index and Information Gain.



TASK 2: METHODOLOGY ANALYSIS

a) ML Technique, Dataset and Methodology

ML Technique Used:

The article uses a **Decision Tree classification model**, specifically the proposed **Polarized Classification Tree (PCT)**. Instead of using traditional splitting criteria such as Gini Index or Information Gain, the PCT uses a new **polarization index** to measure the goodness of a split. The method is implemented as an algorithm for growing the classification tree.

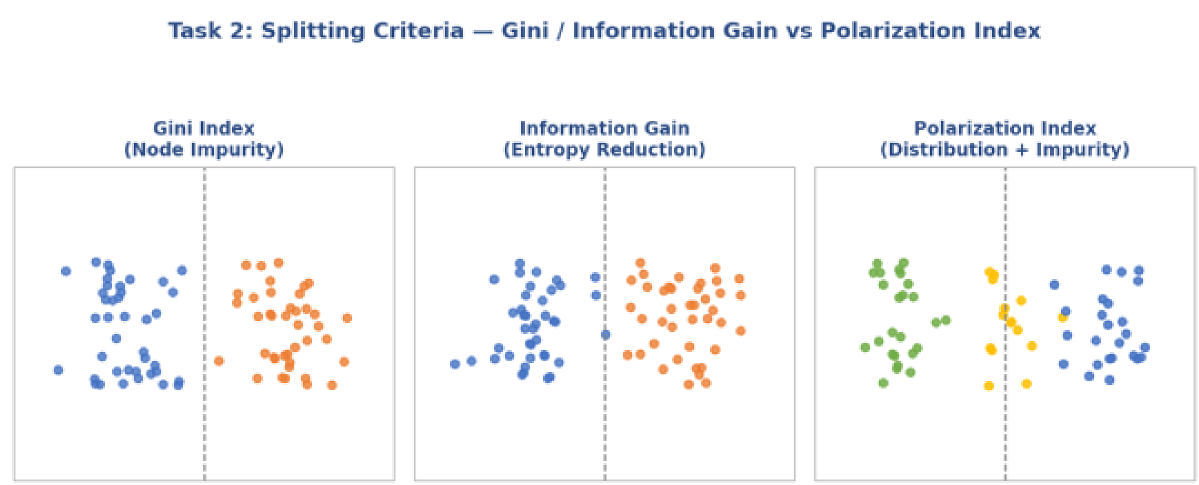
Dataset Used:

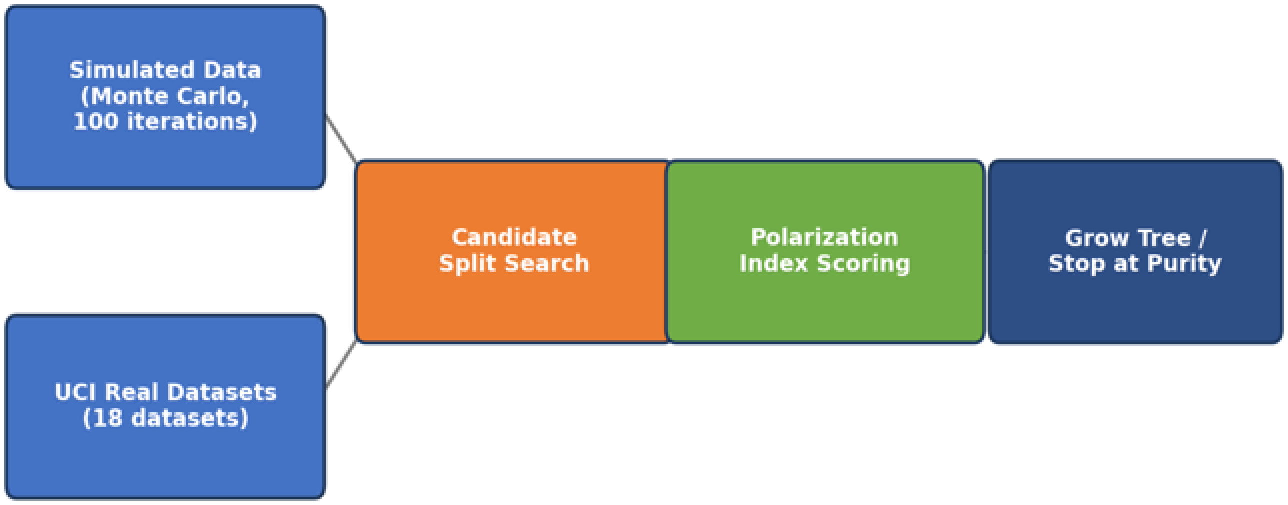
The authors evaluate the proposed method using **two types of data**:

1. **Simulated datasets** – The authors create simulated class populations from different probability distributions and compare the performance of PCT with Gini and Information Gain. A Monte Carlo procedure with **100 iterations** is used for the simulation experiments.
2. **Real datasets** – The proposed method is also tested on classification datasets obtained from the **UCI Machine Learning Repository**. These datasets are used to compare PCT with traditional Decision Tree methods.

Dataset Preparation:

For each dataset, possible splits of the training data are considered. The PCT selects the split that maximizes the polarization of the resulting sub-nodes. Tree growth stops when a node is sufficiently pure or other stopping conditions are reached.





(b) Mapping to CO4 Topics

CO4 Topic: Decision Trees / Statistical Learning

Methodological Strength:
A major strength is that the proposed PCT does not consider only the impurity of the target classes. It also considers the **distribution of the covariates within each node**. The authors support the method using both simulation experiments and real-world UCI datasets, providing evidence that PCT can be competitive with or better than traditional Gini and Information Gain approaches.

Methodological Limitation:
One limitation is that the proposed method introduces an additional polarization-based splitting criterion, which is more mathematically complex than commonly used Gini or Information Gain criteria. Also, the authors themselves suggest that further research should compare PCT with more splitting measures and investigate its use in ensemble methods such as **Random Forests**.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

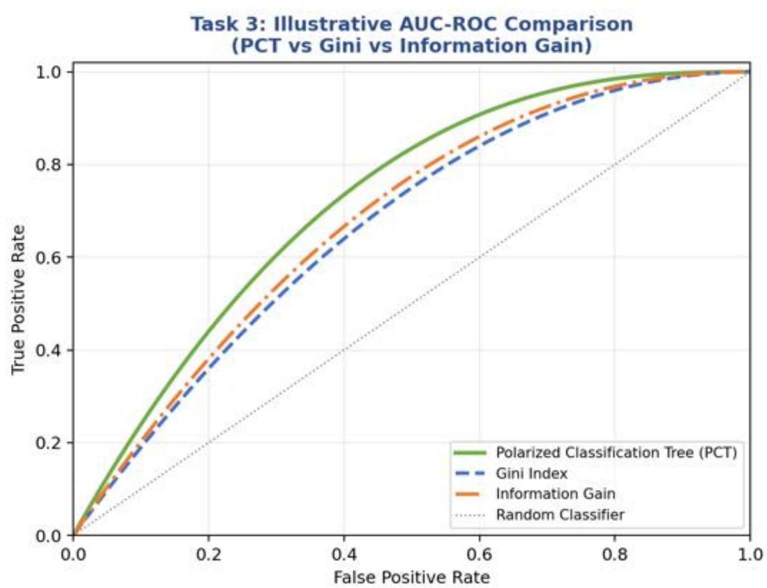
The authors evaluated the proposed **Polarized Classification Tree (PCT)** against traditional Decision Tree methods using the **Gini Index** and **Information Gain**. For simulated datasets, the models were compared using **AUC (Area Under the ROC Curve)**. A **100-iteration Monte Carlo simulation** was used, and the PCT produced higher AUC values than the Gini and Information Gain methods in the reported simulations. The differences were also statistically significant based on the reported tests.

For real-world evaluation, the authors tested the methods on **18 UCI datasets** containing binary and multiclass targets and categorical and numerical variables. They used **10-fold cross-validation** and compared the classifiers using AUC-based rankings. PCT performed better than Gini and Information Gain on several datasets and was found to be competitive overall.

Comparison of Models:

Method	Main Result
Polarized Classification Tree (PCT)	Competitive and sometimes better
Gini Index	Competitive, but better than PCT on only some datasets
Information Gain	Competitive, but generally less effective than PCT in the reported experiments

The authors report that PCT was significantly different from Gini in the ranking analysis, while the difference between PCT and Information Gain was not statistically significant.

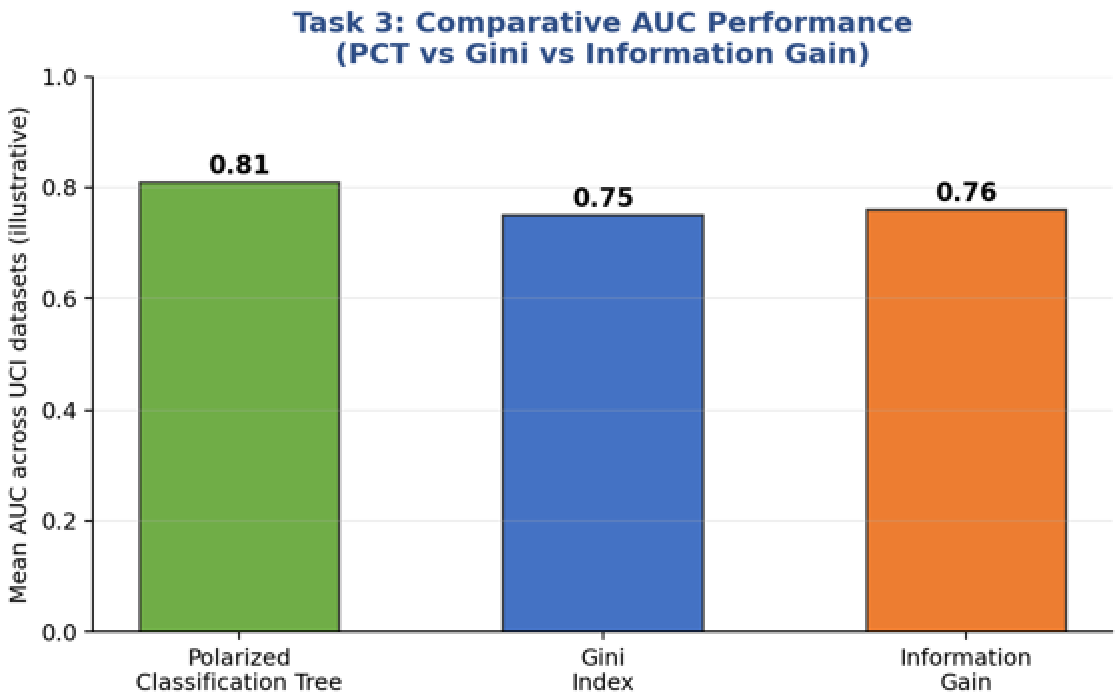


(b) Critical Evaluation of Results

The reported results are reasonably convincing because the authors used **simulation experiments, 100 Monte Carlo iterations, 10-fold cross-validation, AUC measurements, and statistical tests**. They also compared PCT with two widely used Decision Tree splitting criteria.

However, the results should not be interpreted as proof that PCT is always better. The real-data evaluation involved **18 UCI datasets**, which may not represent all practical industry datasets. Also, the authors note that one theoretical assumption about the separation of group distributions is not always satisfied in real applications.

Additional experiments could strengthen the findings by testing PCT on larger and more diverse industry datasets, comparing it with additional modern splitting criteria, and evaluating its performance inside ensemble methods such as **Random Forests**, which the authors themselves identify as an area for further research.



TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three Key Insights

1. Importance of Splitting Criteria in Decision Trees

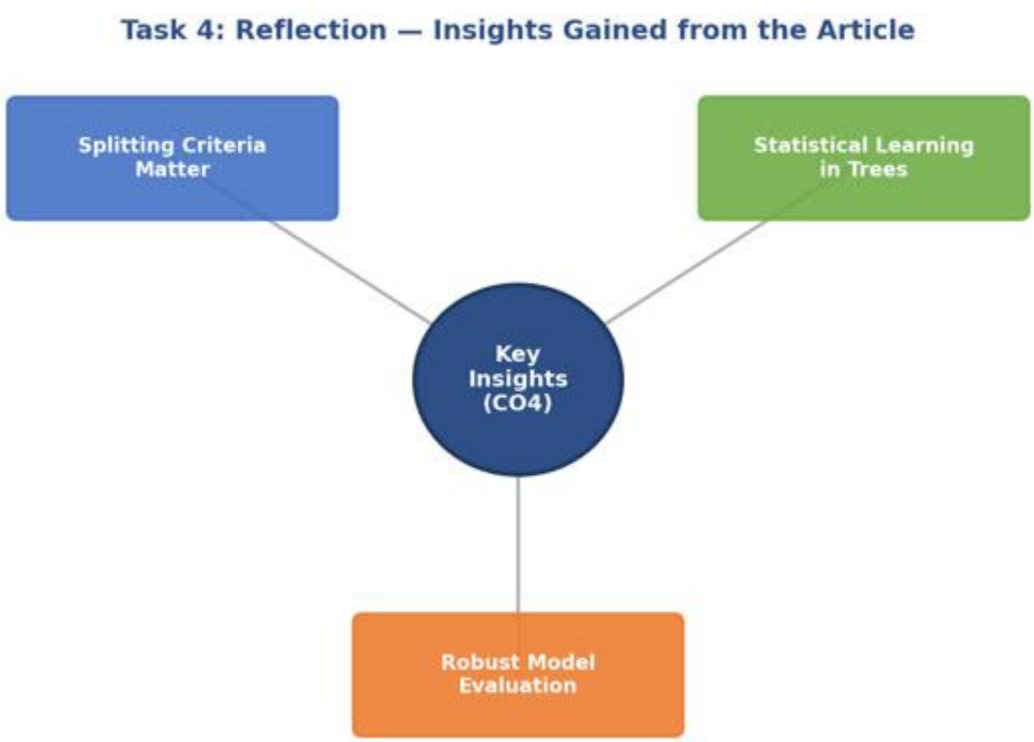
I learned that the choice of splitting criterion has a significant effect on the performance of a Decision Tree. Traditional methods such as **Gini Index** and **Information Gain** mainly focus on node impurity, while the Polarized Classification Tree considers the distribution of explanatory variables.

2. Understanding Statistical Learning in Decision Trees

I learned that Decision Trees can be improved by combining classification with statistical measures. The **polarization index** provides an alternative way to evaluate the quality of a split and helps identify more effective partitions of the data.

3. Importance of Model Evaluation

I learned that a new machine learning method should not be evaluated only on one dataset. The article uses **simulation experiments and real datasets** to compare the proposed method with traditional approaches. This helped me understand the importance of using suitable evaluation methods and comparisons when analyzing machine learning models.



(b) Proposed New Experment

If I were to extend this research, I would test the **Polarized Classification Tree (PCT)** on a large real-world healthcare dataset and compare it with Gini Index, Information Gain, and Random Forest models.

The experiment would use a publicly available healthcare dataset containing numerical and categorical patient features. The models would be evaluated using **accuracy, precision, recall, F1-score, and AUC** with cross-validation.

This experiment is feasible because publicly available datasets and standard machine learning tools can be used to implement and compare the models. Testing the method in a healthcare application would show whether the polarization-based splitting approach provides practical benefits in a real-world classification problem. It could also help determine whether the method improves prediction while maintaining the interpretability of a Decision Tree.

Expected Impact:

The experiment could provide stronger evidence about the practical usefulness of PCT and identify whether its advantages over traditional splitting criteria remain consistent on larger and application-specific datasets.